Higher Order C's and Renormalizability

Terminology

We have seen in our study of loop corrections that we can absorb the divergent terms into the coefficients of the Lagrangian (counter-terms).

If this is true for all higher-orders, we have a theory that is renormalizable.

In fact, if there are a finite number of terms we can still add to the Lagrangian to achieve renormalizability, it is still renormalizable.

If the number of such terms is infinite then the theory is said to be non-renormalizable.

Despite the ∞ parameters, it is still possible to make meaningful predictions at energies below some ultraviolet cut-off.

Conditions for Renormalizability

Say we have a scalar field theory in d spacetime dimensions:

$$\mathcal{L} = -\frac{1}{2} Z_\varphi \partial^\mu \varphi \partial_\mu \varphi - \frac{1}{2} Z_m m^2 \varphi^2 - \sum_{n=3}^{\infty} \frac{1}{n!} Z_n g_n \varphi^n$$

If a Feynman diagram has: $E \rightarrow$ external lines, $I \rightarrow$ internal lines, $L \rightarrow$ closed loops, $V_n \rightarrow$ vertices connecting n lines. The question we ask is, Do the momentum integrals for this diagram diverge?

Remember each closed loop adds a factor $d^d\ell$ and an internal propagator: $1/p^2+m^2$

It is therefore reasonable to suggest that if the number of ℓ's in the numerator is greater than the denominator, the integral will diverge. We can use this to define a superficial degree of divergence:

$D \equiv dL - 2I$; Divergence: $D \geq 0$

Notice how this formula fails to account for dimensional regularization and associated process.

The tree-diagram is also a contributor with mass dimension $[g_E]$. All diagrams must have the same dimensions since they all contribute to the same scattering amplitude.

$\therefore [g_E] = [\text{diagram}] = dL - 2I + \sum_{n=3}^{\infty} V_n [g_n]$ $\longrightarrow$ mass dimension of any diagram is given by the sum of its components.

closed loops are in d dimensions.

With a bit of rearranging we get:

$D = [g_E] - \sum_{n=3}^{\infty} V_n [g_n]$ $\longrightarrow$ The Final Formula.

From the above formula, it is evident that the theory is renormalizable for $[g_n] < 0$.

We also know that:

$$[g_n] = d - \frac{1}{2}n(d-2) \quad ; \quad [g_n] < 0 \text{ if } n > \frac{2d}{d-2}$$

These conditions limit us to only max^m φ^4 in 4 dimensions, φ^3 in 6 dimensions.

This criterion is applicable to more complicated theories as well:

"A theory is non renormalizable if any coefficient of any term in the lagrangian has negative mass dimension."

Q: This does not make sense to me, what if we have a theory with the mass dimension:

$[g_E] + [g_F] - [g_g] + [g_n]$ corresponding to each term. How then is the above statement valid if $[g_g] \gg [g_n]$?

A: Remember that we are talking about mass dimensions, these DO NOT add up and hence the situation described above is WRONG.

We need a detailed study of momentum integrals in Feynman Diagrams to prove this.

Now that we have locked in the negative mass dimensions, lets focus on what happens when its positive / 0.

The only tricky situation here is that posed by the tree diagrams with $[g_E] \geq 0$. This however barely an inconvenience and we can absorb the divergences into Z_E.

This same discussion can be applied to the propagator where the divergences are absorbed into the counter-terms.

Now consider the following diagrams:

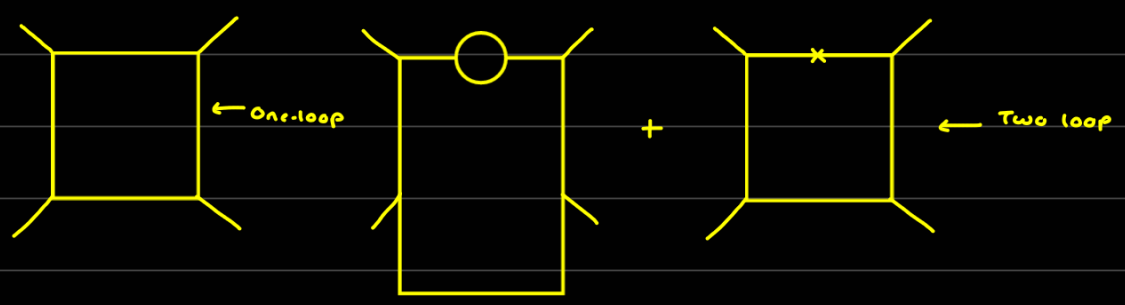

Diagram 2 has that bubble which makes it divergent, This is called a divergent subdiagram. However, you can see that the counter-term on the right cancels the divergence.

You can also see how the —O— part of the diagram is just the loop part of the $\Pi(k^2)$ where the counter terms eliminate the divergences.

* My inference from this is just that if we have a finite number of different subdiagrams (divergent), we will then have a finite number of counter-terms and hence renormalizability.

Thus, theories with all positive/zero coupling mass dimensions are renormalizable.

This applies exactly only to spin-zero and spin ½ fields. spin 1 fields are renormalizable for $d=4$ iff the spin fields are associated with a gauge symmetry

Spin 1 fields are never renormalizable for $d \geq 4$.